

An Optimal Method for Locating and Sizing Distributed Generations Considering Critical Infrastructures Interconnection for Urban Distribution Network Restoration

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Abstract—The urban distribution network will separated from the large-scale centralized power system when disasters occur, causing a series of internal random faults and resulting in serious shortage of energy supply. Distributed Generations (DGs) with renewable energy is significant to restore critical loads and the energy supplying of important interconnected infrastructures to maintain the operation of cities. This paper proposes an optimization method for locating and sizing DGs: (1) Monte Carlo simulation to set faulted scenarios of the urban distribution network after disasters. (2) A mixed integer second-order cone program (MISOCP) is proposed to minimize the number and capacity of DGs, considering the constraints of DGs, water supply network, distribution network and water-electricity coupling relationship. (3) Representative planning solutions are selected from the aspects of power supply reliability, DGs configuration economy, and faulted lines severity. (4) It is proved that the optimization method is effective, multiple-selectivity and forward-looking for the loads restoration and emergency response after disasters, so as to promote the sustainable development of cities with DGs' low-carbon and flexibility.

Keywords—distributed generations with renewable energy sources, locating and sizing, urban distribution network restoration, water-electricity coupling

I. INTRODUCTION

Natural disasters can cause extensive social, economic and even environmental damage [1]. Blackouts due to extreme events leads to disconnection between the distribution network and main network easily. Using Distributed Generations (DGs) to restore critical loads improve the resilience of urban distribution network. The paper proposed an optimal scheduling model for the optimal power grid configuration from the beginning of the faults to the end of the island to maximize the expected recovery load [2-3]. However, the above researches that about the restoration of urban

distribution network focus on giving priority to power loads, while ignoring the energy needs of other critical infrastructures.

There are lifeline facilities in cities that are closely coupled with each other such as electricity, water and gas, and their normal operation is directly related to the vital interests of residents [4]. It is crucial to establish the energy interconnection model of infrastructure for the sustainable development of cities. In this regard, study the capacity optimization of pumped storage station under given wind power output [5]. A steady-state analysis model of integrated electricity and gas systems was proposed to consider the interaction between the two systems from the perspective of energy flow [6]. Reference [7] studied the potential ability of water grid to provide services of renewable resource management demand response for power grid by using water pumps as controllable loads. A joint optimal dispatch method of distribution network and water distribution network was proposed for reducing the joint operation cost of coupled systems and improving the consumption level of distributed generation by optimizing the operation model of water pumps [8]. In [9], the energy elasticity model of water distribution system was proposed and the paper related it to the power regulation range of distribution network nodes. The above researches concerning system coupling provides an important basis for considering the interconnection of key infrastructures for urban distribution network restoration.

To sum up, the studies about locating and sizing DGs are generally concerned from the perspective of economic optimization, which can provide some ideas for the optimization of distribution network restoration, but cannot be directly applied. Therefore, this paper studies the sites selection and capacity determination optimization method of DGs considering critical infrastructures interconnection for urban distribution network restoration.

The remainder of this paper is organized as follows. Section 2 describes the settings of system scenarios and problem statement, while Section 3 formulates a relaxed

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MISOCP for DGs site selection and capacity determination. General simulation results are shown in Section 4. Analysis and discussion of DGs planning solutions with different objectives are detailed in Section 5 before moving into Section 6, which conclude the whole study.

II. PROBLEM STATEMENT

The structure of restoration system is shown in Fig. 1. Power grid includes a variety of distributed power and load. Distributed generation includes PV (Photovoltaic) and WT (Wind Turbine), and there are ordinary loads and coupling loads. And the blackouts due to extreme events cause the external and internal faults of the urban distribution system.

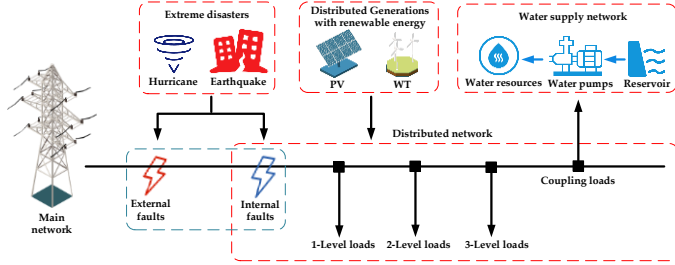


Fig. 1. Using DGs for service restoration after extreme events

In this paper, the details of the system scenarios are set as follows: (1) Natural disasters such as hurricanes and earthquakes caused internal random faults. (2) Using DGs with renewable energy [10], is an effective method for loads restoration in the urban distribution network. (3) The interconnection of critical infrastructures is the premise of different types of energy coupling such as electricity and water. Water pumps are added into distribution network as loads to realize the connection of electricity and water in this paper. (4) The level of loads in the urban distribution network can be divided into 1-Level (L1), 2-Level (L2), and 3-Level (L3) that is differentiated by the load importance degree. Besides, the level of infrastructures coupling loads is L1.

III. RESEARCH METHOD

Based on the statement above, a mixed integer second-order cone program (MISOCP) is formulated to solve the problem of locating and sizing DGs.

A. Objective Functions

In order to minimize the economic cost of DGs configuration on the premise of achieving the basic restoration goal, the objective is expressed as follows where f_1 means the number of DGs and f_2 means the capacity of DGs:

$$\min f = f_1 + f_2 \quad (1)$$

$$\begin{cases} f_1 = \sum_{j \in N_e} \beta_j \\ f_2 = \sum_{j \in N_e} \beta_j P_{DG,j} + \sum_{j \in N_e} \beta_j Q_{DG,j} \end{cases} \quad (2)$$

where, N_e is the set of nodes in the distribution system. β_j is a 0-1 variable, indicating the installation status of DGs. If a DG is located at a node, β_j is 1, 0 otherwise. $P_{DG,j}$, $Q_{DG,j}$ represent the actual total active and reactive power capacity of DGs, respectively.

B. Constraints

The constraints of MISOCP include the DGs constraints, water supply network constraints and distribution network constraints.

1) *DGs Constraints*: (3)~(5) represent the minimum limitation of DGs' number, actual active and reactive power capacity. (6) means the DGs are supposed to restore at least L1 and L2 power loads and coupling loads.

$$\sum_{j \in N_e} \beta_j \geq 1 \quad (3)$$

$$P_{DG,\min} \leq \sum_{j \in N_e} \beta_j P_{DG,j} \quad (4)$$

$$Q_{DG,\min} \leq \sum_{j \in N_e} \beta_j Q_{DG,j} \quad (5)$$

$$\sum_{j \in N_e} (W_j \times x_j) \geq \gamma \quad (6)$$

where, $P_{DG,\min}$, $Q_{DG,\min}$ represent the lowest limitation of the total active and reactive power capacity of DGs, respectively. W_j is the weight coefficient of a load. x_j is a 0-1 variable, which indicates the load status. If a load is recovered, x_j is 1, otherwise is 0. γ means the sum of L1 and L2 loads weights, including coupling loads.

2) Water Supply Network Constraints

a) *Water Supply Operation Constraints*: (7) denotes the flow balance of the water supply network. (8) represents the change of water head between adjacent nodes. (9) denotes the pipe head loss expressed by Darcy-Weisbach formula [11] while (10) denotes the coefficient in the formula. (11)~(12) represent the limits of each pipe flow and pressure head of each node separately.

$$\sum_{j \in E_w} Q_{ij} = q_i, \forall i \in N_w \quad (7)$$

$$H_i + h_i = (H_j + h_j) + \Delta h_{ij}, \forall i \in N_w, \forall i \rightarrow j \in E_w \quad (8)$$

$$\Delta h_{ij} = f_{ij} Q_{ij}^2, \forall i \rightarrow j \in E_w \quad (9)$$

$$f_{ij} = \frac{c_{ij} l_{ij}}{2 d_{ij} s_{ij}^2 g}, \forall i \rightarrow j \in E_w \quad (10)$$

$$Q_{ij,\min} \leq Q_{ij} \leq Q_{ij,\max}, \forall i \rightarrow j \in E_w \quad (11)$$

$$H_{i,\min} \leq H_i, \forall i \in N_w \quad (12)$$

where, i, j represent an arbitrary node in the water supply network, respectively. N_w and E_w are the set of nodes and branches. Q_{ij} and q_i are the water flow and demand from node i to node j . H_i and h_i represent the pressure head and altitude of node i . Δh_{ij} is the head loss of branch ij . f_{ij} is the coefficient of the branch head loss formula. c_{ij} is the friction coefficient of the segment. l_{ij} , d_{ij} and s_{ij} represent the length, diameter and cross-sectional area of pipe ij , respectively.

b) *Pumps Energy Supply Constraints*: (13) represents the active power consumed by the pumps in a period of time while (14) indicates the head loss of pumps. The energy consumption of water pumps is critical to connect the power grid with the water network.

$$P_p = - \frac{\Delta h_p Q_p \Delta t \rho g}{\eta_p} \quad (13)$$

$$\Delta h_p = \omega_p^2 [h_p - r_p (\frac{Q_p}{\omega_p})^{\eta_p}] \quad (14)$$

where, p_p represents a pump active power. Δh_p , Q_p are the head loss and water flow of a pump. ρ is the fluid density. η mean a pump efficiency. ω indicates the nominal value of a pump speed relative to its rated speed and h_p , r_p , η_p are parameters of pump.

3) Distribution Network Constraints

a) *Power Flow Constraints*: (15) represents the power flow constraint. (17) is the deformation of the Ohm's law and lines transmission power.

$$\sum_{k:j \rightarrow k} S_{jk} = \sum_{i:i \rightarrow j} (S_{ij} - z_{ij} l_{ij}) + S_{j,DG}, \forall j \in N_e \quad (15)$$

$$S_j = S_{j,DG} - S_{j,load} - S_{j,p}, \forall j \in N_e \quad (16)$$

$$-(1 - \alpha_{jk})M \leq v_j - v_k - 2\text{Re}(S_{jk} z_{jk}^*) + |z_{jk}|^2 l_{jk} \leq (1 - \alpha_{jk})M, \forall j \rightarrow k \in E_e \quad (17)$$

$$|S_{jk}|^2 = v_j l_{jk}, \forall j \rightarrow k \in E_e \quad (18)$$

(18) is a non-convex constraint. Thus, a relaxation method is used to change it to a convex constraint to obtain a global optimal solution [12].

$$v_j l_{jk} \geq |S_{jk}|^2, \forall j \rightarrow k \in E_e \quad (19)$$

where, i , j and k represent an arbitrary node in the system. N_e and E_e denote the set of nodes and branches. S_{jk} and S_{ij} mean the complex power while z_{ij} and z_{jk} mean the impedance of branch ij . l_{ij} and l_{jk} are the square of branches current. $S_{j,DG}$ and $S_{j,load}$ are the actual output of DG and load power that connected to node j , respectively.

b) *Security Constraints*: (20)~(21) represents the steady-state voltage constraints of each node and steady-state current constraints of each line, respectively.

$$V_{j,min} \leq V_j \leq V_{j,max}, \forall j \in N_e \quad (20)$$

$$I_{jk,min} \leq I_{jk} \leq I_{jk,max}, \forall j \rightarrow k \in E_e \quad (21)$$

where, $V_{j,max}$ and $V_{j,min}$ represent the upper and lower bounds of node voltage V_j . $I_{jk,max}$ and $I_{jk,min}$ represent the maximum and minimum limits of line transmission current I_{jk} .

C. Work Flow

The work flow of locating and sizing DGs, which include four steps are shown in Fig. 2.

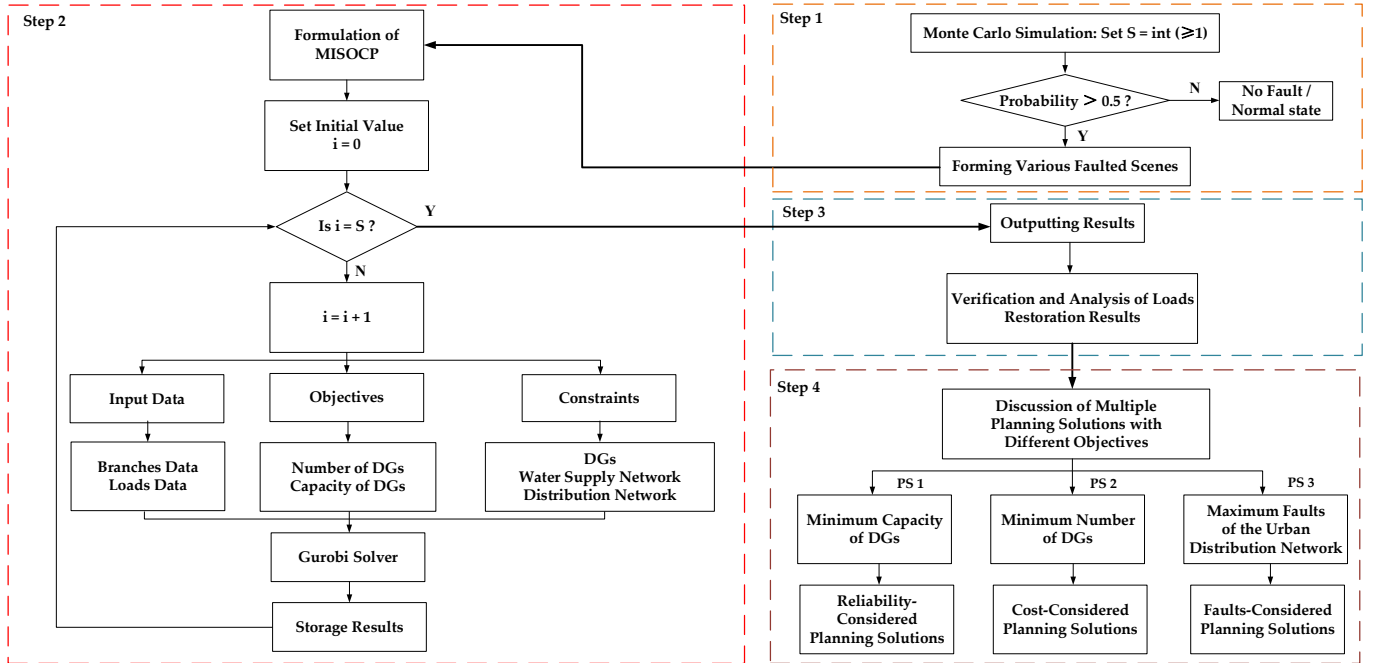


Fig. 2. The work flow of DGs sites selection and capacity determination

- *Step1*: Setting scenarios: Using Monte Carlo simulates different random internal faulted scenarios with a total of S in the urban distribution network.
- *Step2*: Formulating MISOCP: According to optimization objectives, proposing the objective functions and constraints to form MISOCP.
- *Step3*: Outputting the optimization results: Forming various DGs planning solutions that all meet the basic objectives by Gurobi solvers in different scenarios.

- *Step4*: Discussion on respective solutions: Selecting different DGs planning solutions with power supply reliability, DGs configuration economy, and lines faults severity to analyze and discuss.

IV. CASE STUDY

The effectiveness of the proposed model is validated in a modified 37-node PDS, in which the original and modified systems are shown in Fig. 3. This model has been programmed in MATLAB R2019a and the MISOCP model was optimized with Gurobi solver.

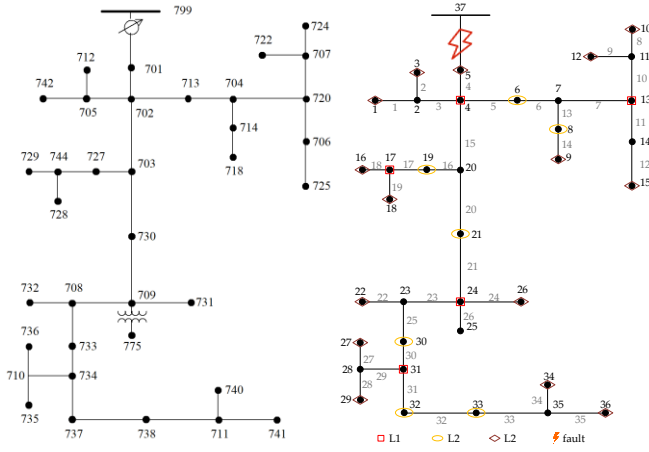


Fig. 3. The original and modified IEEE 37-node systems

A. Parameter Setting

It is assumed that the extreme events happened and the faulted lines are 37-5 (the connection line between the urban distribution network and main network) and the rest of internal lines status that will be discussed by Monte Carlo simulation.

In this system, the nodes with loads are arranged from large to small according to the number of connected branches. When the nodes are connected with the most branches, the level is set as L1 while it's set as L3 if the nodes are connected with the least branches, and the others is L2. The nodes with the largest number of connected branches are connected to pumps, whose level is set as L1. The W1, W2, W3 of L1, L2, L3 are 100, 10, 0.2, respectively. There are 5 L1 loads, 7 L2 loads and 15 L3 loads, with a total of 2783.94 kW+1201 kVar in this system.

B. Random Faults Simulation

There are random faults of the internal lines in the urban distribution network due to external factors such as tower collapse, wind and rain damage, etc. This paper uses Monte Carlo to simulate the random failure probability that is represents by $P_i(j)$. For the probability results of simulation, 0.5 is assumed critical point of the faults occurrence.

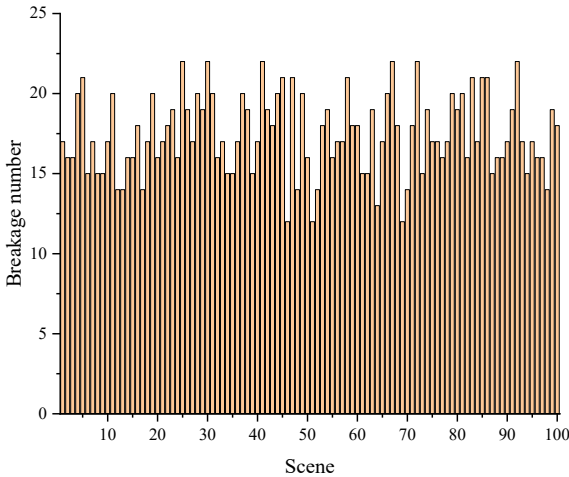


Fig. 4. The results of number of actual lines faults by Monte Carlo

Actually, setting the number of $S=100$, $E_c=35$ (the number of the rest of internal lines). The number of faulted lines (define the number as N_{fi}) that the probability greater than 0.5 under every scenarios are shown in Fig. 4. It can be seen that the probability distribution is uniform with the faulted probability = 0.5 as the reference plane. The maximum N_{fi} is 22, corresponding to 6 scenarios while the minimum is 12 with 3 scenarios. The results of Fig. 4 prove the randomness and applicability of Monte Carlo simulation.

C. General Results

Based on the requirement of objective functions and various constraints, the general results meet the need of restoring power loss loads, that is, the restoration number of weighted loads $\geq 570 = \sum_{j \in L1, L2} \omega_j \times x_j$ (in this case).

Further, representative DGs planning solutions are discussed from different perspectives including reliability-considered, cost-considered and faults-considered.

V. RESULTS ANALYSIS AND SOLUTIONS DISCUSSION

A. Analysis of General Results

Under the requirements of the proposed MISOCP, the critical power loads and coupling loads have been restored considering the system losses, which means Constraint (6) is satisfied. The detailed active and reactive power output of each DG under every faulted scenarios are shown in Fig. 5.

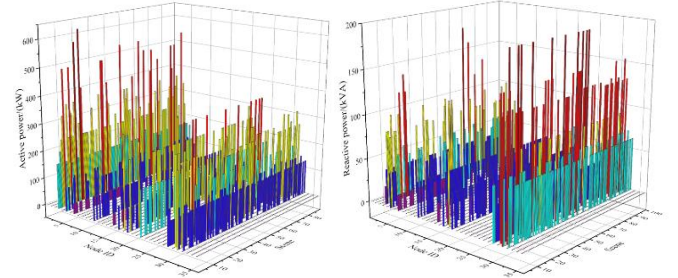


Fig. 5. Statistics data (Pg and Qg) of DGs in each scenario

The minimum configuration capacity of DGs is 1099.94 kW. It is further proved that the all DGs planning solutions by Monte Carlo simulation can satisfy the restore objective of urban distribution network. Specially, it is important to note that under the objective of restoring critical loads, there are slight differences in the total capacity of DGs due to different lines losses in various scenarios, which is further reflected in the different number and location of DGs installation.

From the above results, there are some interesting phenomena. For example, the DGs configuration capacity is the minimum when it is just equal to the sum of power of L1 and L2 loads, including coupling loads. But, the number of DGs installation is the most. However, the result of DGs capacity that is the most when the number of DGs is the least.

B. Discussion of Representative Planning Solutions of DGs

Based on the overall results of Section 4, paper further discuss from different point of view. DGs planning focuses on locating and sizing, in which considers the reliability, cost and

faults severity to form various alternative solutions and their discussion are as follows.

1) Reliability-Considered Planning Solutions

As reliability, according to the urban distribution network requirements, loads restoration can be considered quickly and stably after extreme events. In general results, it can be found the solutions with the least capacity configuration have the most number of DGs. In this condition, DGs are set beside each important load, which can ensure the surrounding DGs can supply power reliably when loads lose power. The most reliable solutions among results are detailed in Table I.

TABLE I. THE RELIABILITY-CONSIDERED PLANNING SOLUTIONS OF DGs

Solutions	Total Capacity of DGs (kW)	Number of DGs	Installation Node	Single Capacity of DGs (kW)
15	1099.94	12	4,6,8,13,17,19,21,24,30,31,32,33	164.97,85,38,85,42,42,85,164.97,85,42,140,126
100	1099.94	12	4,6,8,13,17,19,21,24,30,31,32,33	164.97,85,38,85,42,42,85,164.97,85,42,140,126

There are two solutions that are configured exactly the same, thus, choosing solution 15 for further analysis. As shown in Fig. 6, there are 16 faults in the distribution network. DGs are installed near each restoration load, in which the maximum capacity is 164.97 kW while the minimum is 38 kW. The number of isolated electric islands is 11, among which DG8 and DG9 are connected to form an island due to there are no faults between line 22~25.

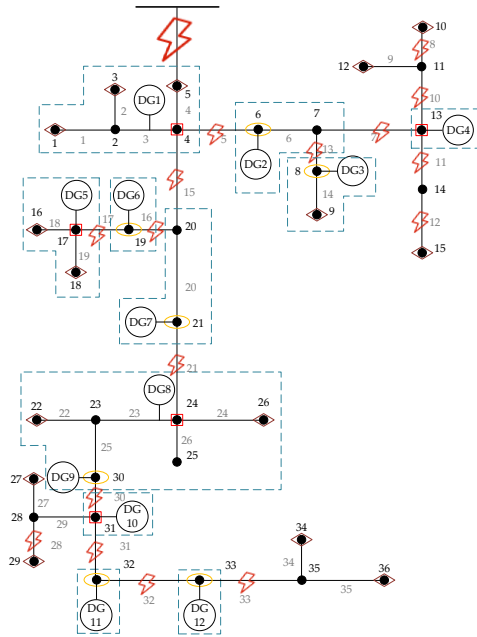


Fig. 6. Configuration result with reliability-considered (solution 15).

This solutions show that DGs has the highest reliability in all results when the total capacity of DGs with 1099.94 kW is the minimum that is equal to the total power of L1 and L2 loads including coupling loads, because it can achieve

independent restoration for each load and there are the least lines losses in the system.

2) Cost-Considered Planning Solutions

As same as reliability, the cost is also an important factor for DGs configuration. On the one hand, it's important to consider the cost of per unit capacity of renewable energy [13]. On the other hand, considering the limitation and availability of urban land, the number of DGs should be as small as possible.

In the general results, it can be found that the solutions of the least number of DGs with the most capacity. There are only 5 DGs are located in different sites in this system. The solutions are detailed in TABLE II.

TABLE II. THE COST-CONSIDERED PLANNING SOLUTIONS OF DGs

Solutions	Capacity of DGs (kW)	Number	Installation Node	Single Capacity of DGs (kW)
12	1101.75	5	4,7,17,20,30	250,123.08,42,292.40,394.27

As shown in Fig. 7, the faulted zones are divided into 5 isolated electric islands by 5 DGs, and each island includes a L1-critical load.

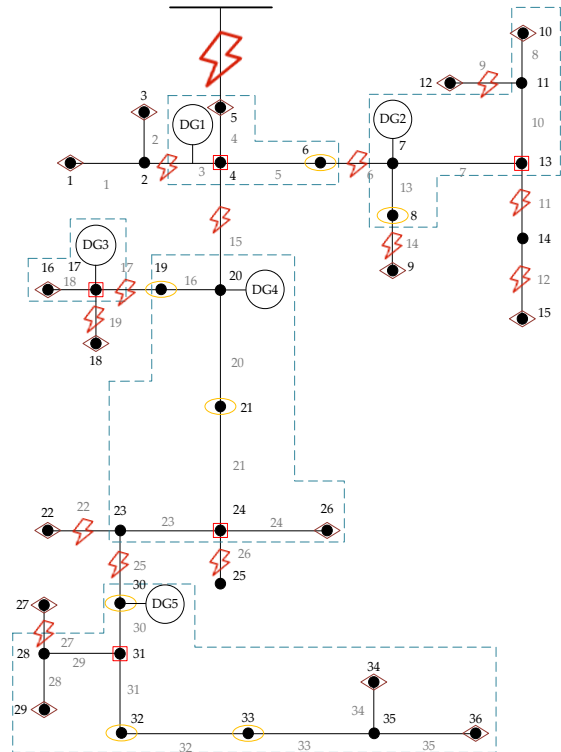


Fig. 7. Configuration result with cost-considered (solution 12)

The solutions indicate that the maximum capacity of DGs is 1101.75 kW, which is different from 1099.94 kW (the minimum capacity) due to each DG restores at least 2 critical loads of different nodes at the same time. So, there are higher lines losses in the system. From the perspective of urban sustainability, the cost is the lowest among the general results.

3) Faults-Severity-Considered Planning Solutions

Faults-severity is measured by the number of disconnected branches in the urban distribution network. To further illustrate the universal applicability of this proposed method, 6 solutions with the most serious faults are analyzed from 2 perspectives. It can be found all solutions that are detailed in TABLE III. to meet the basic objectives.

TABLE III. THE FAULTS-SEVERITY-CONSIDERED PLANNING SOLUTIONS OF DGs

Solutions	Capacity of DGs (kW)	Number	Installation Node	Single Capacity of DGs (kW)
25	1100.08	9	4,8,13,19,24,3 0,31,32,33	335.10,38,85,84.01,164. 97,85,42,140,126
30	1100.27	8	4,8,13,17,24,3 0,31,33,	377.15,38,85,42,164,97, 85,182.15,126
41	1099.95	11	4,6,8,13,19,21, 24,30,31,32,33	164.97,85,38,85,84.01,8 5,164.97,85,42,140,126
67	1099.98	11	4,8,13,17,19,2 1,24,30,31,32, 33	250.01,38,85,42,42,85,1 64.97,85,42,140,126
72	1100.15	9	4,6,8,13,19,21, 30,31,33	164.97,85,38,85,84.01,2 50.02,85,182.15,126
92	1099.97	10	4,6,8,13,17,21, 24,30,32,33	206.99,85,38,85,42,85,1 64.97,127.01,140,126

As shown in Fig. 8, from the most configuration capacity (the least cost) point of view, solution 30 makes the least number and the lowest cost of DGs among 6 solutions. And as shown in Fig. 8, considering the least capacity (the most reliability), the DGs number is the most that can make sure the power supply reliability in solution 41 but its cost is highest among 6 solutions.

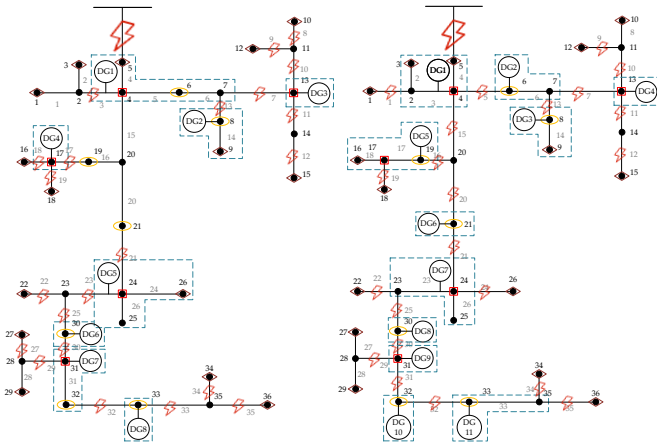


Fig. 8. Configuration results with faults-severity-considered (solution 30 and solution 41)

The configuration capacity is between 1099.97~1100.27 kW, and the number of DGs is between 8 and 11. Although the number of disconnected lines is the same (22 faulted lines), the configuration of DGs is different due to different disconnection positions.

VI. CONCLUSION

An optimal method for locating and sizing Distributed Generations (DGs) with renewable energy is proposed for

restoring the urban distribution network, while respecting the interconnection of critical infrastructures (water-electricity coupling). A relaxed MISOCP is used to model this problem while considering various constraints of this model. The results indicate that multiple DGs planning solutions with different objectives can be obtained effectively and the most suitable solution can be selected according to various strategy focus to further promoting the sustainable development of cities while meeting the emergency response and energy demand.

In future work, more resource types, the operational characteristics of other energy networks except water network, and their coupling relationship with distribution network should be studied.

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